

```
sage 6.8.rc1: bash - Konsole
sreedevis@redevi-X550LC:~/sage-6.8.rc1$ ./sage
SageMath Version 6.8.rc1, Release Date: 2015-07-22
Type "notebook()" for the browser-based notebook interface.
Type "help()" for help.

Warning: this is a prerelease version, and it may be unstable.

sage: matrix([[1,2],[3,4]])
(1 2)
(3 4)
sage: matrix([[4,7],[0,1],[0,4,5]])
(4 7 3)
(0 1 0)
(0 4 5)
sage: 
```

Figure 5: Constructing a matrix

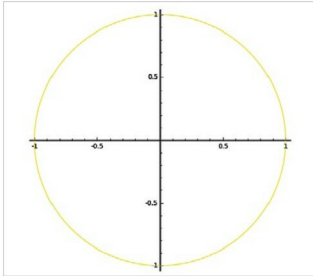


Figure 6: Drawing a unit circle with 0,0 as centre

```
cd sage-6.8.rc1
```

The `ls` command will list the files and folders inside the Sage directory. Please go through the `README.txt` for further information.

The build process

You need to give the `make` command for the build process to happen; however, it does take some time.

When the build gets over, to start using Sage, type the following command from Sage's home directory:

```
./sage
```

Figures 6 and 7 illustrate a few examples where Sage has been used to get solutions to mathematical problems.

You can use the Sage reference manual to look at more such examples.

Advantages

SageMath can be used in research, teaching, to learn various courses, etc. Everyone can download and use it. You can take a demo from a class and publish it, so that students

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sage: plot(x^2, (0, 1))
(0, 0)
(1, 1)
sage: 
```

Figure 7: Command

can look at it after the class. Mathematica has the 'Player' application but with SageMath, students can easily alter the demo for themselves — something that isn't possible with the former. Since many users will be familiar with Python, Sage encourages students to discover bugs, and to try and fix them.

SageMath allows network transparency. For instance, if SageMath is running on a fast machine, you will get the benefit of that machine even while accessing Sage from somewhere else. Since Mathematica's support for Linux is not very good, SageMath is a true open source alternative to it.

Contributing to Sage

Many people, including high school students, undergraduates, graduate students, mathematicians, scientists and software engineers contribute to the Sage project. Sage has a very welcoming developer community with a lot of options for contributing to it, like finding a bug and reporting it, fixing a bug, implementing a new function or creating a new class. If you are not familiar with programming, you can still contribute to Sage by writing technical tutorials for a mathematical topic, translating existing documentation to a new language, etc.

You can ask questions about contributing to Sage in its IRC channel `#sagemath` on `freenode.net` and in its mailing lists (<https://groups.google.com/forum/#!forum/sage-devel>).

Do refer to the following link for more information:

<http://doc.sagemath.org/html/en/developer/>

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